

TechStore-MVC-Python

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A sophisticated Inventory Management System built with Python, utilizing the **Model-View-Controller (MVC)** architectural pattern. This project demonstrates clean code principles, dependency injection, and robust database transaction handling for secure sales processing.



Key Features

- **MVC Architecture:** Explicit separation of concerns— `models.py` for data logic, `views.py` for UI, and `controllers.py` for orchestration.
- **Atomic Transactions:** Uses SQL transaction blocks (commit/rollback) in the `StoreModel` to prevent inventory discrepancies.
- **Dependency Injection:** The `DatabaseProvider` is injected into the model, making the system modular and easier to test.
- **Automated Schema Setup:** A robust `setup.py` script that verifies the database environment, creates tables, and inserts initial dummy data.
- **Modern GUI:** A clean Tkinter interface featuring a `Treeview` for inventory display and automated UI refreshes after transactions.



Tech Stack

- **Core:** Python 3
- **GUI Framework:** Tkinter
- **Database:** MySQL (with `mysql-connector-python`)

File Overview

- `main.py` : The entry point that initializes all components and starts the application.
- `models.py` : Contains the `Product` dataclass and `StoreModel` for CRUD operations.
- `views.py` : Manages the graphical layout and user input.
- `controllers.py` : Handles events from the view and updates the model.
- `database.py` : Provides a centralized way to acquire database connections.
- `setup.py` : Handles environment preparation and initial database seeding.

Installation & Usage

1. Prerequisites

Ensure a MySQL server is active and accessible (default configuration is `root` with no password).

2. Install Requirements

```
pip install mysql-connector-python
```

3. Run Database Setup

Run the following command to prepare your local environment:

```
python setup.py
```

4. Start the System

```
python main.py
```

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